

Two Simulations for Mystery Tokens

Unit 3 · Topic 3.14 · THE PROBLEM

Suppose a company claims 10,000 mystery tokens are 50% red, 30% teal, and 20% gold. An SRS of 80 has the shown counts.

Leo: “Mix and redraw these same 80 tokens without replacement; recompute χ^2 for 1,000 trials.”

Amara: “For each trial, generate 80 independent colors with probabilities 0.50, 0.30, and 0.20, then recompute χ^2 .” In Amara’s 1,000 trials, 186 statistics are at least the observed value.

Token sample

Color	Claim	Observed
Red	0.50	32
Teal	0.30	30
Gold	0.20	18

FIRST TAKE — one sentence before you work the parts

Which plan can produce a different number of red tokens on the next trial? Explain in one sentence.

- DOK 1** (a) Find the three expected color counts from the company’s percentages.
- DOK 2** (b) The observed statistic is $\chi^2 = 3.35$. Check the red contribution $(32 - 40)^2/40$, then compute the simulation proportion $186/1,000$.
- DOK 3** (c) In 3–4 sentences, choose a simulation and justify what varies from trial to trial. Use the simulated proportion to decide whether the result is unusual at a 5 percent cutoff, and explain why this does not prove the company’s claim.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.